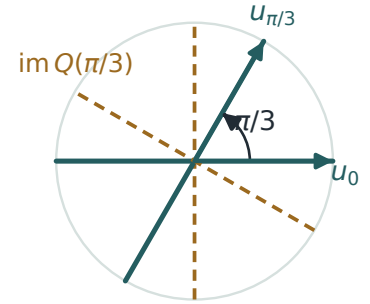
A rotating line and its complementary packet

$u_\theta = (\cos\theta, \sin\theta)$, $P(\theta) = u_\theta u_\theta^\top$, $Q(\theta) = I - P(\theta)$.



Solid: P packet Dashed: Q packet

Canonical continuous transport

$$K = [P', P] = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$U'(\theta) = KU(\theta), \quad U(0) = I$$

$$U(\theta) = R(\theta)$$

$$S = \cos\delta\, R(\delta), \qquad S^\top S = \cos^2\delta\, I$$

For $|\delta| < \pi/2$: polar$(S) = R(\delta)$ exactly.

Steps N	Each increment	Ordered product
2	$\pi/6$	$R(\pi/3)$
4	$\pi/12$	$R(\pi/3)$
8	$\pi/24$	$R(\pi/3)$

Every mesh product is exact in this finite model.
The general theorem concerns convergence for broader projector paths.